MATH 152 **QUIZ 4** Name Solutions

Mrs. Bonny Tighe 7.2, 7.3

25 points TA_____Wed 9/27/2023

1. Evaluate the integrals.

a) $\int t\,(\sin^2 t)\,dt$ $\quad u = t \quad dv = \sin^2 t\,dt = \frac{1}{2} - \frac{1}{2}\cos 2t\,dt$

$du = dt \quad v = \frac{1}{2}t - \frac{1}{4}\sin 2t$

$= t\sin^2 t - \int\left(\frac{1}{2}t - \frac{1}{4}\sin 2t\right)dt$

$= t\sin^2 t - \frac{1}{4}t^2 + \frac{1}{8}\cos 2t + C$

$= t\sin^2 t - \frac{1}{4}t^2 - \frac{1}{8}\cos 2t + C$

b) $\int \sec^3(3x)\tan^3(3x)\,dx$

$\int \sec^2(3x)\tan^2(3x)\underbrace{\sec(3x)\tan(3x)}\,dx$

$u = \sec(3x)$

$\tan^2(3x) = \sec^2(3x) - 1 \quad du = 3\sec(3x)\tan(3x)\,dx$

$\int \sec^2(3x)(\sec^2(3x) - 1)\sec(3x)\tan(3x)\,dx$

$\int u^2(u^2 - 1)\cancel{\sec(3x)}\cancel{\tan(3x)}\left(\frac{du}{3\cancel{\sec(3x)}\cancel{\tan(3x)}}\right)$

$\frac{1}{3}\int(u^4 - u^2)\,du = \frac{1}{3}\cdot\frac{1}{5}u^5 - \frac{1}{3}\cdot\frac{1}{3}u^3 + C$

$\frac{1}{15}\sec^5(3x) - \frac{1}{9}\sec^3(3x) + C$

c) $\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}dx$

$x = 2\sin\theta$

$dx = 2\cos\theta\, d\theta$

$1 = 2\sin\theta$

$1/2 = \sin\theta, \quad \theta = \pi/6$

$\sqrt{3} = 2\sin\theta$

$\sqrt{3}/2 = \sin\theta, \quad \theta = \pi/3$

$$\int_{\pi/6}^{\pi/3} \frac{\sqrt{4-4\sin^2\theta}\,(2\cos\theta\, d\theta)}{4\sin^2\theta} = \int_{\pi/6}^{\pi/3} \frac{\sqrt{4(1-\sin^2\theta)}\,2\cos\theta\, d\theta}{4\sin^2\theta}$$

$$\int_{\pi/6}^{\pi/3} \frac{2\cos\theta \cdot 2\cos\theta\, d\theta}{4\sin^2\theta} = \int_{\pi/6}^{\pi/3} \frac{\cos^2\theta}{\sin^2\theta}d\theta = \int_{\pi/6}^{\pi/3} \cot^2\theta\, d\theta$$

$$\int_{\pi/6}^{\pi/3} (\csc^2\theta - 1)\,d\theta = -\cot\theta - \theta \Big]_{\pi/6}^{\pi/3}$$

$$\left(-\cot\pi/3 - \pi/3\right) - \left(-\cot\pi/6 - \pi/6\right) = -\sqrt{3}/3 - \pi/3 + \sqrt{3} + \pi/6 \text{ or}$$

$$2\sqrt{3}/3 - \pi/6$$

d) $\int \frac{1}{\sqrt{4x^2+9}}dx$

$2x = 3\tan\theta$

$2x/3 = \tan\theta$

$x = 3/2 \tan\theta$

$dx = 3/2 \sec^2\theta\, d\theta$

$$\int \frac{1}{\sqrt{9\tan^2\theta+1}}\left(3/2 \sec^2\theta\, d\theta\right) = \int \frac{3/2 \sec^2\theta\, d\theta}{\sqrt{9(1+\tan^2\theta)}} = \int \frac{3/2 \sec^2\theta\, d\theta}{3\sec\theta}$$

$$\int \frac{1}{2}\sec\theta\, d\theta = \frac{1}{2}\ln\left|\sec\theta + \tan\theta\right| + C$$

$$\int \frac{1}{\sqrt{4x^2+9}}dx = \frac{1}{2}\ln\left|\frac{\sqrt{4x^2+9}}{3} + \frac{2x}{3}\right| + C$$

Complete the Square

$x^2+2x+1-8-1$

$(x+1)^2-9$

e) $\int \frac{\sqrt{x^2+2x-8}}{x+1}\,dx$

let $(x+1) = 3\sec\theta$

$dx = 3\sec\theta\tan\theta\,d\theta$

$$\int \frac{\sqrt{(x+1)^2-9}}{x+1}\,dx = \int \frac{\sqrt{9\sec^2\theta-9}}{3\sec\theta}(3\sec\theta\tan\theta)\,d\theta$$

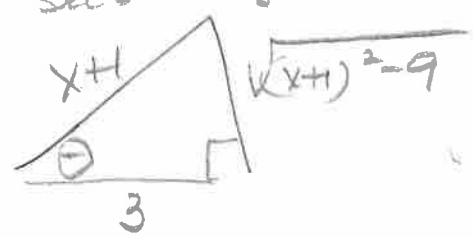
$$\int \frac{\sqrt{9(\sec^2\theta-1)}\,\cancel{3\sec\theta}\tan\theta\,d\theta}{\cancel{3\sec\theta}} = \int 3\tan\theta\tan\theta\,d\theta$$

$$3\int \tan^2\theta\,d\theta = 3\int(\sec^2\theta-1)\,d\theta =$$

$$3\tan\theta - 3\theta + C$$

$$3\left(\frac{\sqrt{(x+1)^2-9}}{3}\right) - 3\left(\sec^{-1}\left(\frac{x+1}{3}\right)\right) + C$$

$\sec\theta = \frac{x+1}{3}$



f) $\int \tan^4(2x)\,dx$ $= \int \tan^2(2x)\tan^2(2x)\,dx$

$$\int((\sec^2(2x)-1)\tan^2(2x)\,dx =$$

$$\int \tan^2(2x)\sec^2(2x)\,dx - \int \tan^2(2x)\,dx$$

$u = \tan(2x)$

$du = 2\sec^2(2x)\,dx$

$\frac{du}{2\sec^2(2x)} = dx$

$$\int u^2\cancel{\sec^2(2x)}\left(\frac{du}{2\cancel{\sec^2(2x)}}\right) = \int \frac{1}{2}u^2\,du = \frac{1}{6}u^3 + C$$

$$\frac{1}{6}\tan^3(2x) + C$$

MATH 152 **QUIZ 4A** Name Solutions
Mrs. Bonny Tighe 7.2, 7.3
25 points TA_____Wed 9/27/2023

1. Evaluate the integrals.

a) $\int \tan^4(2x)\,dx = \int \tan^2(2x)\tan^2(2x)\,dx$

$u = \tan(2x)$
$du = 2\sec^2(2x)\,dx$
$\frac{du}{2\sec^2(2x)} = dx$

$\int \tan^2(2x)(\sec^2(2x)-1)\,dx =$

$\int \tan^2(2x)\sec^2(2x) - \frac{\tan^2(2x)}{1}\,dx$

$\int u^2 \cancel{\sec^2(2x)}\left(\frac{du}{2\cancel{\sec^2(2x)}}\right)dx - \int(\sec^2(2x)-1)\,dx$

$\frac{1}{2}\int u^2\,du - \left[\frac{1}{2}\tan(2x) - x\right] + C$

$\frac{1}{2}\cdot\frac{1}{3}u^3 - \frac{1}{2}\tan(2x) + x + C$

$\frac{1}{6}\tan^3(2x) - \frac{1}{2}\tan(2x) + x + C$

b) $\int \sec^4 x \tan^3 x\,dx$ — there are 2 ways

#1 $u = \tan x$ $du = \sec^2 x\,dx$

or #2 $u = \sec x$ $du = \sec x \tan x\,dx$

#1 $\int \sec^2 x \tan^3 x \sec^2 x\,dx$

$\int (1+\tan^2 x)\tan^3 x \sec^2 x\,dx$

$\int u^3 + u^5\,du =$

$\frac{1}{4}\tan^4 x + \frac{1}{6}\tan^6 x + C$

#2 $\int \sec^3 x \tan^2 x \sec x \tan x\,dx =$

$u = \sec x$
$du = \sec x \tan x\,dx$

$\int \sec^3 x(\sec^2 x - 1)\sec x \tan x\,dx$

$\int (\sec^5 x - \sec^3 x)\cancel{\sec x}\tan x\,dx$

$\int (u^5 - u^3)\,du = \frac{1}{6}u^6 - \frac{1}{4}u^4 + C$

$\frac{1}{6}\sec^6 x - \frac{1}{4}\sec^4 x + C$

c) $\int t\cos^2 t\,dt$

$u = t$ $\quad dv = \cos^2 t\,dt = \frac{1}{2} + \frac{1}{2}\cos 2t$

$du = dt$ $\quad v = \frac{1}{2}t + \frac{1}{4}\sin 2t$

$= t\left(\frac{1}{2}t + \frac{1}{4}\sin 2t\right) - \int\left(\frac{1}{2}t + \frac{1}{4}\sin(2t)\right)dt$

$= \frac{1}{2}t^2 + \frac{t}{4}\sin 2t - \frac{1}{4}t^2 + \frac{1}{8}\cos(2t) + C$

d) $\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}\,dx$

$x = 2\sin\theta$

$dx = 2\cos\theta\,d\theta$

$\sqrt{3} = 2\sin\theta,\ \sin\theta = \sqrt{3}/2 \quad \theta = \pi/3$

$1 = 2\sin\theta,\ \sin\theta = 1/2 \quad \theta = \pi/6$

$$\int_{\pi/6}^{\pi/3} \frac{\sqrt{4-4\sin^2\theta}}{4\sin^2\theta}(2\cos\theta)\,d\theta$$

$$\int_{\pi/6}^{\pi/3} \frac{\sqrt{4(1-\sin^2\theta)}}{4\sin^2\theta}(2\cos\theta)\,d\theta = \int_{\pi/6}^{\pi/3} \frac{2\cos\theta(2\cos\theta)\,d\theta}{4\sin^2\theta}$$

$$\int_{\pi/6}^{\pi/3} \frac{\cos^2\theta}{\sin^2\theta}\,d\theta = \int_{\pi/6}^{\pi/3} \cot^2\theta\,d\theta =$$

$$\int_{\pi/6}^{\pi/3} (\csc^2\theta - 1)\,d\theta = -\cot\theta - \theta\Big]_{\pi/6}^{\pi/3}$$

$$(-\cot\pi/3 - \pi/3) - (-\cot\pi/6 - \pi/6)$$

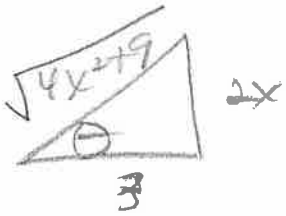
$$\left(-\sqrt{3}/3 - \pi/3\right) - \left(-\sqrt{3} - \pi/6\right) = -\frac{\sqrt{3}}{3} - \frac{\pi}{3} + \sqrt{3} + \pi/6$$

aside $\sqrt{3} - \sqrt{3}/3 = \frac{3\sqrt{3}}{3} - \frac{\sqrt{3}}{3} = \frac{2\sqrt{3}}{3}$

$$= -\pi/6 + 2\sqrt{3}/3$$

e) $\int \frac{1}{x\sqrt{4x^2+9}}dx$

$2x = 3\tan\theta$ $\quad \tan\theta = 2x/3$



$x = \frac{3}{2}\tan\theta$

$dx = \frac{3}{2}\sec^2\theta\, d\theta$

$$\int \frac{1}{\frac{3}{2}\tan\theta\sqrt{9\tan^2\theta+9}}\left(\frac{3}{2}\right)\sec^2\theta\, d\theta$$

$$\int \frac{1}{\tan\theta\sqrt{9(\tan^2\theta+1)}}(\sec^2\theta\, d\theta) = \int \frac{1}{\tan\theta(3\sec\theta)}(\sec^2\theta\, d\theta)$$

$$\int \frac{1}{3}\frac{\sec\theta}{\tan\theta}d\theta = \frac{1}{3}\int \frac{\frac{1}{\cos\theta}}{\frac{\sin\theta}{\cos\theta}}d\theta = \frac{1}{3}\int \frac{1}{\cancel{\cos\theta}}\cdot\frac{\cancel{\cos\theta}}{\sin\theta}d\theta$$

$$\frac{1}{3}\int \frac{1}{\sin\theta}d\theta = \frac{1}{3}\int \csc\theta\, d\theta = \frac{1}{3}\ln|\csc\theta - \cot\theta| + C$$

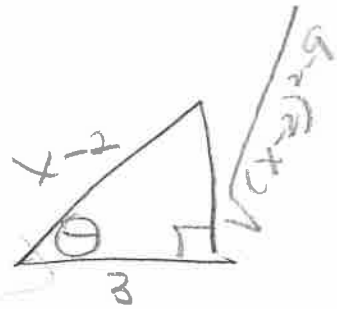
$$\frac{1}{3}\ln\left|\frac{\sqrt{4x^2+9}}{2x} - \frac{3}{2x}\right| + C$$

f) $\int \frac{\sqrt{x^2-4x-5}}{x-2}dx$

complete the square

$x^2-4x+4-5-4$

$(x-2)^2-9 =$



$(x-2) = 3\sec\theta$

$dx = 3\sec\theta\tan\theta\, d\theta$

$$\int \frac{\sqrt{(x-2)^2-9}}{x-2}dx$$

$$\int \frac{\sqrt{9\sec^2\theta-9}}{3\sec\theta}(3\sec\theta\tan\theta)\, d\theta$$

$$\int \sqrt{9(\sec^2\theta-1)}\tan\theta\, d\theta$$

$$\int 3\tan\theta\tan\theta\, d\theta = 3\int \tan^2\theta\, d\theta$$

$$3\int(\sec^2\theta-1)d\theta = 3(\tan\theta-\theta)$$

$$3\left(\frac{\sqrt{(x-2)^2-9}}{3}\right) - 3\sec^{-1}\left(\frac{x-2}{3}\right) + C$$